PRELIMINARY SPECIFICATION

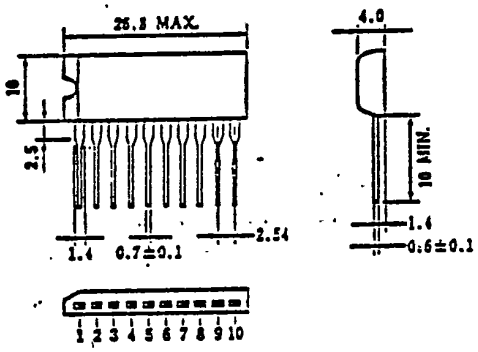
NEC ELECTRON DEVICE

MOS FIELD EFFECT POWER TRANSISTOR ARRAY

μPA1520H

FAST SWITCHING N-CHANNEL SILICON POWER MOS FET ARRAY

PACKAGE DIMENSIONS
in millimeters



2,4,6,8:Gate
3,5,7,9:Drain
1,10:Source

FEATURES

- Suitable for switching power supplies, actuater controls,and pulse circuits
- Low RDS(on)
- No second breakdown

ABSOLUTE MAXIMUM RATINGS (Ta=25 ℃)

Drain to Source Voltage	VDSS	30V
Gate to Source Voltage	VGSS	±20V
Continuous Drain Current	ID(DC)	± 2A
Total Power Dissipation	PT	3.5W
Total Power Dissipation	PT*	28W
Channel Temperature	Tch	150℃
Storage Temperature	Tstg	-55~150 ℃

*Tc=25℃

ELECTRICAL CHARACTERISTICS (Ta=25℃)

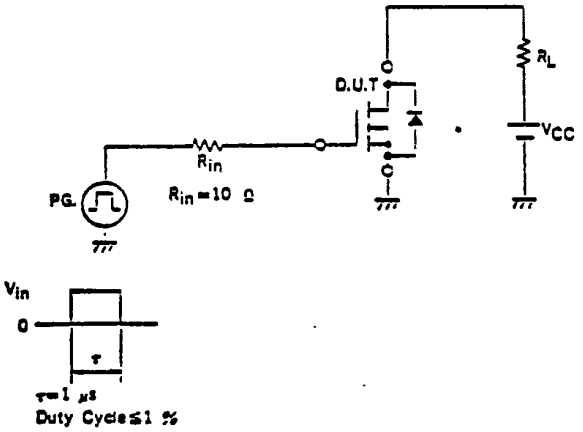
CHARACTERISTIC	SYMBOL	MIN.	TYP.	MAX.	UNIT	TEST CONDITIONS
Drain Leakage Current	IDSS			10	μA	VDS= 30V, VGS=0
Gate to Source Leakage Current	IGSS			100	nA	VGS=±20V,VDS=0
Gate to Source Cutoff Voltage	VGS(off)	1.0		2.5	V	VDS=10V, ID=1mA
Forward Transfer Admittance	|yfs|	1.0			S	VDS=10V, ID=1A
Drain to Source On-State Resistance	RDS(on)		0.11	0.17	Ω	VGS=10V, ID=1A
			0.20	0.25	Ω	VGS=4V, ID=0.8A
Input Capacitance	Ciss		500		pF	VDS=10V
Output Capacitance	Coss		120		pF	VGS=0
Reverse Transfer Capacitance	Crss		30		pF	f=1MHz
Turn-On Delay Time	td(on)		10		ns	ID=1A , Vcc=10V
Rise Time	tr		20		ns	VGS(on)=10V
Turn-Off Delay Time	td(off)		80		ns	RL=10 Ω
Fall Time	tf		20		ns	

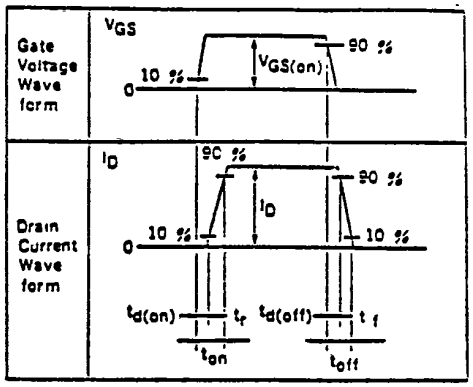
Mar.28.1986 ID-1778

NEC cannot assume any responsibility for any circuits shown or represent that they are free from patent infringement.
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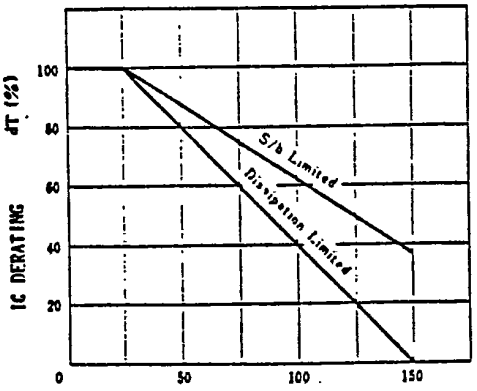
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TURN-ON AND TURN-OFF TIME TEST CIRCUIT



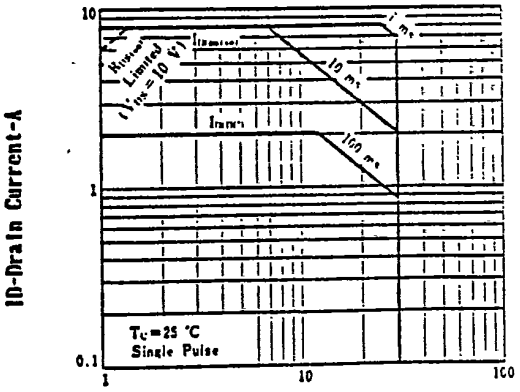


DERATING CURVE OF SAFE OPERATING AREA



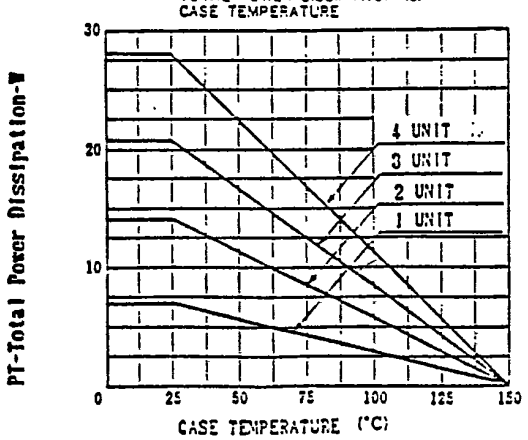
TC-Case Temperature-℃

FORWARD BIAS SAFE OPERATING AREA

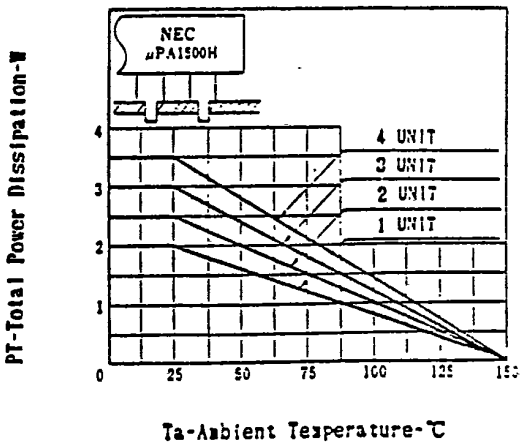


VDS-Drain to Source Voltage-V

TOTAL POWER DISSIPATION vs. CASE TEMPERATURE

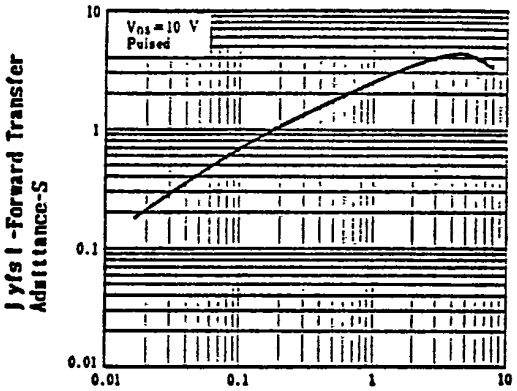


TOTAL POWER DISSIPATION vs. AMBIENT TEMPERATURE

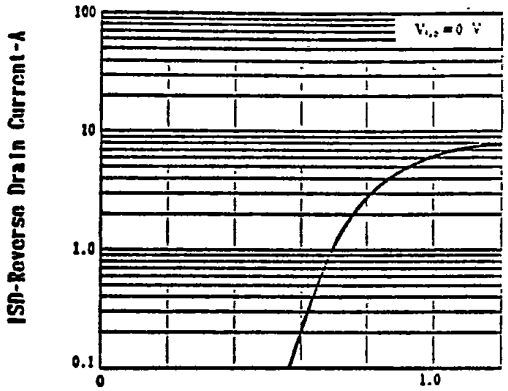


Ta-Ambient Temperature-℃

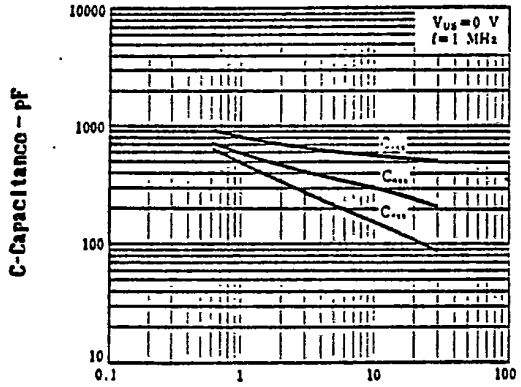
FORWARD TRANSFER ADMITTANCE vs. DRAIN CURRENT



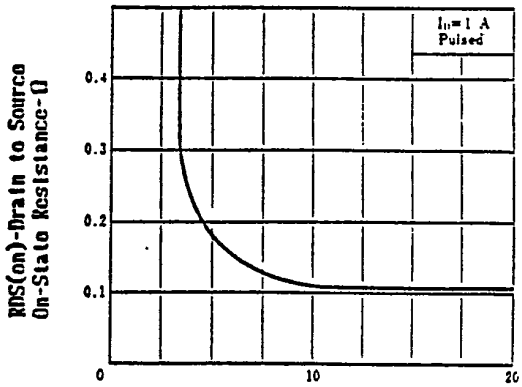
SOURCE TO DRAIN DIODE FORWARD VOLTAGE



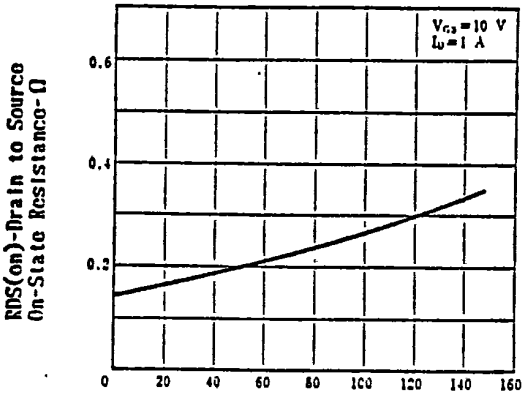
CAPACITANCE vs. DRAIN TO SOURCE VOLTAGE



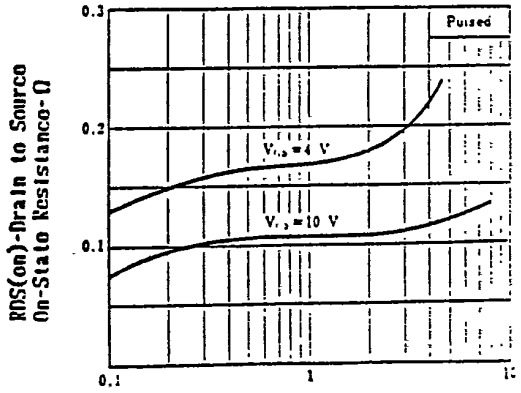
DRAIN TO SOURCE ON-STATE RESISTANCE vs. GATE TO SOURCE VOLTAGE



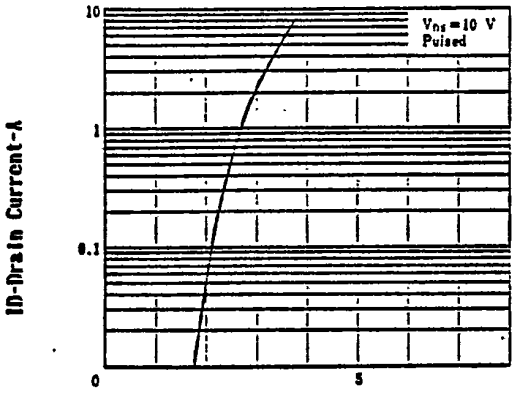
DRAIN TO SOURCE ON-STATE RESISTANCE vs. CHANNEL TEMPERATURE



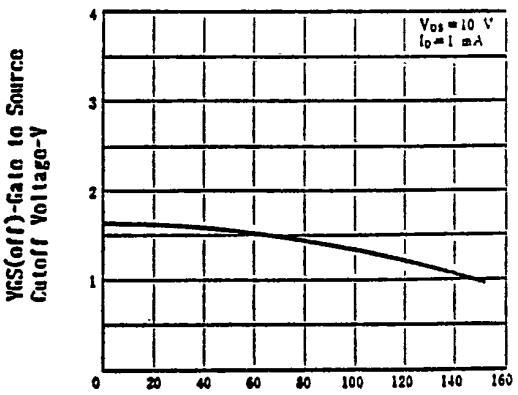
DRAIN TO SOURCE ON-STATE RESISTANCE vs. DRAIN CURRENT



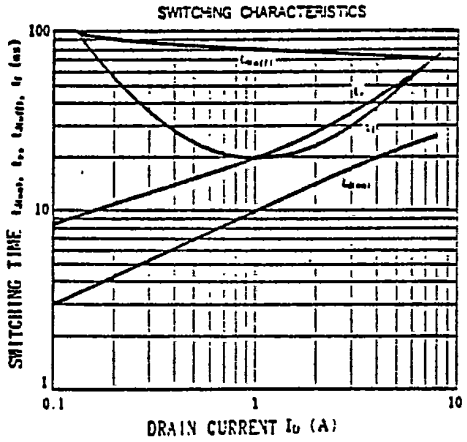
TRANSFER CHARACTERISTICS



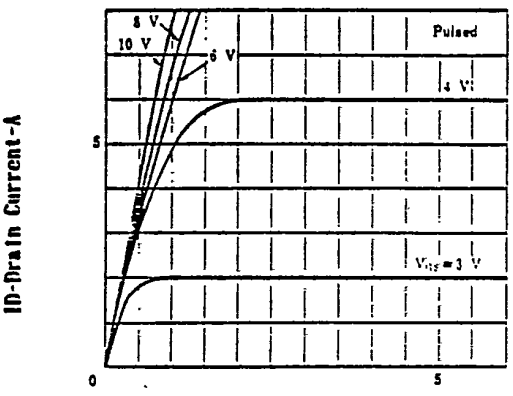
GATE TO SOURCE CUTOFF VOLTAGE vs. CHANNEL TEMPERATURE



SWITCHING CHARACTERISTICS



DRAIN CURRENT vs. DRAIN TO SOURCE VOLTAGE



NEC Corporation

INTERNATIONAL ELECTRON DEVICES DIV.
NEC Building, 33-1, Shiba Gochome
Minato-ku, Tokyo 108, Japan
Tel: Tokyo 454-1111
Telex Address: NECTOK J22686
Cable Address: MICROPHONE TOKYO

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